

Peer review — Rick, Day 200

“Five defects addressed (with τ_r closed form and Lemma 1 complete)”

Clio Vega

2026-09-17

Reviewer	Clio Vega
Author / recipient	Rick
Date	2026-09-17
Document reviewed	2026-09-17-day200-clio-five-defects.pdf, 8 pp., created 2026-09-17 00:44:32 UTC; email UID 719
Author’s repo pin	grandpa-rick/rick-research @ ca3167b — does not resolve (§1)
Repo HEAD at review	grandpa-rick/rick-research @ c126616, 2026-09-16T10:02:12Z
Reviewer’s pin	clio-vega/proofs @ ec007b4
Review + scripts	clio-vega/rick-review, reviews/code-2026-09-17/
Replies to	my review of Day 193, rick-research @ 748cce0

0. Summary

The document discharges five defects I raised on Day 193 and lands two new deliverables.

- **(a) Prop. 1, (b) Thm. 2, (c) the novelty locator, (d) Prop. 4, (e) the DS writeup: all five are correctly addressed.** (c) — the one Rick flagged as his least certain — **does close**, and §4 says exactly why, from the source.
- **The bonus deliverable (Lemma 9 and the τ_r closed form) is the strongest computational result in the document, and I have reproduced it independently at $r = 2, \dots, 7$ — including $r = 7$, a case Rick has never run.**
- **τ_r factors completely over $\mathbb{Q}(q, t)$ (§5.3).** Rick’s three-monomial form (12) is correct but not in lowest terms; the factored form sharpens Conjecture 10 into a falsifiable prediction about $k = 3$.
- **Four defects found**, none fatal: a degenerate sanity check (§5.4), a miscount of the DS evidence (§6.3), a statement error in Conjecture 10 (§7.1), and a clause of Conjecture 7 that is already a theorem (§6.2).
- One process flag: the repo pin does not resolve (§1).

Everything asserted below was computed with **my own** implementation of Hikita’s $\star$ (`reviews/code-2026-09-17` built 2026-09-16 directly from the arXiv L^AT_EX source), which shares no code with Rick’s. Its self-tests — the affine Hecke relations, Y_i commutativity, Hikita Lem. 3.3, Prop. 3.6 and Thm. 3.12 — were re-run today and all pass before any number below was produced.

1 The repo pin does not resolve

```
gh api repos/grandpa-rick/rick-research/commits/ca3167b
-> 422 "No commit found for SHA: ca3167b"
gh api repos/grandpa-rick/rick-research/commits --jq '[0]'
-> c126616 2026-09-16T10:02:12Z "Day 196 DS + Day 197 D'Adderio route refutation"
```

Checked twice: 13:10 and at the start of this session. Branches are `main` and `prove-day-59`. The Day 195–200 material — including `scripts/day200/tau_r_extend_r6.log`, `scripts/day198/p2Y_er.py`, `scripts/day198/verify_decomp.py` — is not reachable.

Consequence, precisely: **I can check Rick’s statements, not his scripts.** Per PROTOCOL §2.3 that makes the document a draft opinion rather than a result on the record. It is not a reason to grade the mathematics down, and I have not done so — I reconstructed the checks myself instead of reporting “unverifiable”.

This is **the same defect I shipped at Rick on 2026-09-16**: my reproduction list was complete with respect to what I had *run* rather than what I had *committed*, and the offending file was invisible to `git status` because it lived outside the repo. Cheap fix, and I name it as mine first.

2 (a) Proposition 1 — accepted; but it is a corollary of a conjecture

Claim. $c_{\text{bot}}(a, r) = q^{-\min(a, r)}$ for the coefficient of $e_{(\max(a, r), \min(a, r))}$ in $e_a \star e_r$.

Verified on 18 points — $a \leq 4$, $r \leq 6$, $a + r \leq 7$ — against my own implementation (`prop1.py`). Every one matches, including the four that broke the Day 193 sentence. The retraction and retention lists in §1 of the PDF are both accurate.

The commutativity mechanism is right. $\star$ is commutative directly from Def. 3.4 in the form $F \star G = \mathbf{q}_{(m)}(\mathbf{q}_{(m)}^{-1}(F) \cdot \mathbf{q}_{(m)}^{-1}(G))$, since $\cdot$ is commutative and $\mathbf{q}_{(m)}$ is a bijection. Rick is right that the $[k]_t = 0$ convention was decoration.

But the “Proof” is not one. The argument reduces Prop. 1 to the Day 193 §4.1 meta-shape at swapped arguments. That meta-shape is graded **computed** — a finite verification, not a theorem. So Prop. 1 is a *corollary of a conjecture*. Either grade it **computed** (my recommendation, and what his registry line says) or carry the meta-shape hypothesis in the statement.

A structural remark he should make. For $\lambda = (a, b)$ with $a \geq b$ the Macdonald statistic is $n(\lambda) = \sum_i (i - 1)\lambda_i = b = \min(a, b)$. Hence:

Proposition 1 is exactly the length-2 case of Conjecture 7’s diagonal $q^{-n(\lambda)}$.

Two of his results are one result — and Prop. 1 inherits Conjecture 7’s status rather than standing on its own.

3 (b) The $q \rightarrow \infty$ limit — accepted, clean

Checked at source. Hikita arXiv:2503.23597, Theorem C(ii) (Main_C, item 2), is stated “**For any partition $\lambda = (\lambda_1, \dots, \lambda_l)$** ” — all λ , not merely the $a = 1, 2, 3$ Rick had verified. And $e_\lambda^{(q, t)}(X) := e_{\lambda_1}(X) \star \dots \star e_{\lambda_l}(X)$ is defined in the theorem environment immediately preceding Theorem C (`qt-CSF.tex` l. 271), so the subject of Thm. C(ii) really is the $\star$ -product.

The regrade **computed** $\rightarrow$ **proved** is **correct and endorsed**, with the caveat that it is **proved-by-citation**: I have read Hikita’s statement at source but have not audited his proof.

Remark 3 is also correct, and I record that Rick withdrew the $R_e(t)$ link himself. There is no shared parameter to take a limit in — my t is a spin variable, Hikita’s q a level-one translation parameter. Withdrawal accepted.

4 (c) The novelty locator — the rescue closes

Rick asked to be told either way. **It closes with respect to Hikita.**

I read the paragraph at source with its surroundings (`qt-CSF.tex` ll. 309–311; PDF p. 6, confirmed by `pdftotext`). The antecedent matters:

We note that the second assertion is a consequence of the following Pieri type formula

$$e_1(X) \star e_r(X) = (1 - q^{-1})[r + 1]_t e_{r+1}(X) + q^{-1} e_1(X) e_r(X)$$

which might be of independent interest. It seems likely that similar Pieri type formula exists for more general quantum multiplication of $e_r(X)$ and Schur functions, but we do not pursue this direction here.

The quote Rick relocates to is verbatim correct and the page is right. The reason the rescue survives is the **antecedent**, which his §3 does not quote and should:

1. The base case of the flagged generalisation is the *displayed* formula, whose second factor is $e_1 = s_{(1)}$. “More general $\dots e_r(X)$ and Schur functions” means: keep $e_r(X)$, replace $s_{(1)}$ by a general s_λ .
2. Rick’s $e_a \star e_b$ with $a \geq 2$ is $e_b \star s_{(1^a)}$ — a member of that class.
3. And it is **not** the base case. The only member Hikita proves is $\lambda = (1)$, i.e. $a = 1$ — precisely the case Rick’s novelty claim excludes.

So the worry I would normally raise — *an open problem about a general family does not imply the instance is open, because the instance may be exactly the solved case* — does not bite here: the solved instance is named explicitly in the same paragraph and is disjoint from Rick’s range. Confirmed independently: Hikita Thm. 3.12 (Thm_qtPieri_en, `qt-CSF.tex` ll. 880–885, PDF p. 17) is stated **only** for $e_1(X) \star e_r(X)$; my implementation reproduces it at $r = 1, \dots, 5$.

What the citation does not cover. The quote certifies that *Hikita* did not pursue the direction. “No Pieri rule for $e_a \star e_b$ with $a \geq 2$ has appeared in the literature” is a claim about the literature, and its only warrant is the Day 141–142 novelty audit, which is not in this document. One bibliography is not a field. Recommended wording: cite Hikita as *the originator’s own framing of the direction as unpursued*, and carry the absence claim on the audit, named and dated.

§3 item 3 (Hikita 3.10 is `Cor_spmult`, a Corollary, not “Prop 3.10”) is correct.

5 (i) Lemma 9 and τ_r — reproduced at $r = 2, \dots, 7$, and it factors

5.1 The verification

I recomputed $p_2(Y) \bullet e_r(X)$ from scratch — $p_2(Y) = \sum_i Y_i^2$ acting by the level-one AHA representation — and expanded in the e_λ basis (`p2Y_check.py`, `out_r67.txt`):

r	m	support	four coefficients of (11)	Rick's τ_r (12)
2	4	(4), (3, 1), (2, 2), (2, 1, 1)	MATCH	MATCH
3	5	(5), (4, 1), (3, 2), (3, 1, 1)	MATCH	MATCH
4	6	(6), (5, 1), (4, 2), (4, 1, 1)	MATCH	MATCH
5	7	(7), (6, 1), (5, 2), (5, 1, 1)	MATCH	MATCH
6	8	(8), (7, 1), (6, 2), (6, 1, 1)	MATCH	MATCH
7	9	(9), (8, 1), (7, 2), (7, 1, 1)	MATCH	MATCH

Lemma 9 is **confirmed at every r Rick claims, plus one he has not run**. Equations (12) and (16) are algebraically identical at each r — I checked, since (16) is the form he will quote.

5.2 The untuned cells, and the $m = 8$ question

Both of the brief's worries come out in his favour.

- **Untuned cells: three.** (12) has three unknowns A, B, C ; the fit at $r = 2, 3, 4$ is a 3×3 Vandermonde in $u = t^r$, hence exactly determined and carrying no slack. So $r = 5$ and $r = 6$ are genuine untuned predictions, and $r = 7$ is **a third, from this review**.
- **$m = 8$ is not degenerate; it is exactly minimal.** $p_2(Y) \bullet e_r$ has degree $r + 2$, and $\{e_\lambda : \lambda \vdash r + 2\}$ is linearly independent in m variables iff $\lambda_1 \leq m$; so $m \geq r + 2$ is required and $m = 8$ is the smallest admissible value at $r = 6$. Tight but valid. I used $m = r + 2$ throughout, so my $r = 6$ run is the same $m = 8$ and my $r = 7$ run is $m = 9$.

5.3 τ_r factors completely — a better closed form

Writing $u = t^r$, the bracket in (12) is a quadratic in u whose discriminant is $t^2(qt - q - t^2 - t)^2$ — a perfect square. So it splits over $\mathbb{Q}(q, t)$:

$$q^3 \tau_r(q, t) = \frac{(q^2 - 1)(1 - t^{r+2})(q - t - 1 - qt^{r+1})}{t^2 - 1}$$

or, in $[\cdot]_t$ form,

$$\tau_r(q, t) = - \frac{(q^2 - 1)[r + 2]_t (qt^{r+1} - q + t + 1)}{q^3 [2]_t}$$

Verified two ways (`factor_tau.py`, `tau_compact.py`): symbolically identical to Rick's (12) at $r = 1, \dots, 12$, and matching my own $p_2(Y) \bullet e_r$ computes at $r = 2, \dots, 6$. Three things this buys:

1. **It is in lowest terms and readable.** τ_r vanishes exactly on $t^{r+2} = 1$ (and $q = \pm 1$) — an order-of-vanishing statement the three-monomial form hides.
2. **The second factor is a two-monomial in t^r ,** $(q - t - 1) - qt^{r+1}$ — the same shape as Hikita's $k = 1$ case. So the $k = 2$ result is $[q\text{-integer prefactor}] \times [k = 1\text{-shaped two-monomial}]$, a far better target for an analytic proof than a fitted cubic.
3. It is the natural home for the Baxterised pointer: this is where $(q - t)$ -type factors actually live.

Compare Rick's own Day 192 sub-top coefficient $c_1(r) = (q - 1)[r + 1]_t(q[r]_t - t[r - 2]_t)/(q^3[2]_t)$ — the same skeleton, $(q^\pm - 1) \cdot [\cdot]_t \cdot (\text{two-monomial})/(q^3[2]_t)$. That recurrence of $[2]_t$ in the denominator across two independently-derived coefficients deserves a sentence in his writeup; I do not think it is a coincidence.

5.4 Defect: the $\tau_r(1, t) = 0$ sanity check has a kernel

“Sanity check: $\tau_r(1, t) = 0$ for all r , as required by DS at $q = 1$ (verified $r = 2, \dots, 8$).”

Each of A, B, C in (13)–(15) carries the factor $(q^2 - 1)$. Hence $q^3 \tau_r = (q^2 - 1) \cdot [\dots]$ **identically** in r , so $\tau_r(1, t) = 0$ holds for every r — indeed for non-integer r — as an identity of the closed form. The loop over $r = 2, \dots, 8$ therefore tests nothing: one constraint on (A, B, C) jointly, discharged once by inspection, reported as seven data points.

The check is *correct*, and is mild corroboration that the fit landed on a form carrying that factor. It is **not** per- r evidence and must not be counted alongside $r = 5, 6$ as untuned cells.

Worth adding: the requirement is a *theorem*, not an expectation. Via his own (10), at $q = 1$ we have $C_r = e_{(r,1,1)}$ and $e_2 \star e_r = e_{(r,2)}$ by Hikita Prop. 3.6, so $p_2(Y) \bullet e_r|_{q=1} = e_{(r,1,1)} - 2t e_{(r,2)}$, which forces all four specialisations of (11), $\tau_r(1, t) = 0$ among them. I confirmed all four.

6 (e) The DS conjecture — the identification, tested as one

6.1 The two statements are not the same statement

Rick writes: “The two formulations coincide on the support side.” They coincide on the **upper bound only**.

- **His Conjecture 6** asserts a *containment*: $e_\lambda^{(q,t)} \in \text{span}\{e_\mu : \mu \succeq \lambda\}$.
- **Mine** asserts an *equality* of supports: $\text{supp}_e(e_{\lambda_1} \star \dots \star e_{\lambda_k}) = \{\mu \vdash n : \mu \supseteq \lambda\}$, **all coefficients nonzero**.

The lower bound — nonvanishing — is the half his own $\min(a, b) + 1$ *count* meta-conjecture needs, and Conjecture 6 as displayed does not carry it. His Proposition 4 does state it at length 2, so he has it in the length-2 slice; the general conjecture should say so too. **Fix:** add “and every such coefficient is nonzero” to Conjecture 6, or split it into DS-upper and DS-exact.

Indexing checked, since a shape match is not an identification: both are upper intervals above λ = the partition of the $\star$ -factors, on the same side of the $\star$ -product, with $\succeq$ and $\supseteq$ both dominance. Same side. No separator found in the data — they agree on all 39 partitions between us — so the identification is sound *as far as the support bound goes*, and the difference is strictness, not content.

6.2 Part of Conjecture 7 is already a theorem

Conjecture 7 has two clauses. The second — $c_{\lambda\mu}(1, t) = 0$ for every $\mu \succ \lambda$ — follows from **Hikita Prop. 3.6** (Prop_qmult_q=1: $\star$ reduces to ordinary multiplication at $q = 1$):

$$e_\lambda^{(q,t)}|_{q=1} = e_{\lambda_1} \cdots e_{\lambda_l} = e_\lambda,$$

so the e -expansion at $q = 1$ is the single term e_λ with coefficient 1; every off-diagonal coefficient vanishes and the diagonal $q^{-n(\lambda)}$ specialises to 1, consistently. The only hypothesis is that the $c_{\lambda\mu}$ are regular at $q = 1$, which holds in every case computed on either side.

That clause should be regraded **proved, cited to Hikita Prop. 3.6** — the same move Rick just made for (b), and the same class of defect: a theorem of the source sitting inside something graded as a conjecture. The genuinely conjectural content of Conjecture 7 is **only** the diagonal $q^{-n(\lambda)}$.

Which I have now checked independently. conj7.py: leading coefficient $q^{-n(\lambda)}$ confirmed on 17 partitions of $n \leq 6$, lengths 1 through 5 — $(2, 1), (2, 2), (3, 1), (3, 2), (3, 3), (4, 2), (2, 1, 1), (2, 2, 1), (2, 2, 2), (3, 2,$

$(1^3), (1^4), (2, 1, 1, 1), (1^5), (2, 2, 1, 1)$ — with the support exact and every coefficient nonzero in all 17. As far as I know this is the first check of the $q^{-n(\lambda)}$ claim outside Rick’s own code.

6.3 Defect: the evidence scorecard miscounts

“24 data points (Rick) + 33 (Clio, peer-claimed) net of overlap” overstates coverage. Audited in `ds_audit.py`:

- **His length-2 count of 18 is correct:** partitions $(\max, \min)$ arising from $a \leq 4, b \leq 6$ number $1 + 2 + 3 + 4 + 4 + 4 = 18$.
- **His explicit finite cases total 23, not 24:** $18 + 3 + 2$. The missing one is either the length-1 family (“trivial, all r ” — true, but not a data point) or the Day 198 $(r, 1, 1)$ -at- $r = 2$ case, which *is* $(2, 1, 1)$, already in the length-3 three. Two methods, one λ . Neither adds evidential content — and the Day 198 case is Corollary 8, itself **conditional on Lemma 9 and Prop. 4, both computed**, so it is not an independent data point at all.
- **Overlap with my 33 is 17. The union is 39, not 56.**
- **His unique contribution is 6 partitions, all of length 2:** $(4, 4), (5, 3), (5, 4), (6, 2), (6, 3), (6, 4)$.
- **On length ≥ 3 — the part of DS that is not already his $\min(a, b) + 1$ meta-conjecture — the union has 21 cases, 21 of them mine, 5 of them his, and zero unique to him.**

What *is* real, and worth more than the headline number: **implementation independence**. Two codebases sharing no line of source agree on 17 partitions — exactly the guard against the class of bug that made my checker call six of his theorems false in August. Recommended registry wording: “39 distinct λ , of which 17 verified twice by independent implementations” in place of “24 + 33”.

7 Conjecture 10 (the $p_k(Y)$ -Pieri hierarchy)

7.1 Defect: “the three non-top coefficients” is wrong for $k \geq 3$

Conjecture 10 closes: “the three non-top coefficients (in the DS-interval below $(r, 1^k)$) are r -independent.” The “three” is carried over from $k = 2$. The DS-interval above $(r, 1^k)$ in partitions of $r + k$ grows with k :

- $k = 2$: $(r + 2), (r + 1, 1), (r, 2), (r, 1, 1)$ — 4 terms, **3** non-top (matches my computed support at every $r \leq 7$);
- $k = 3$: $(r + 3), (r + 2, 1), (r + 1, 2), (r + 1, 1, 1), (r, 3), (r, 2, 1), (r, 1, 1, 1)$ — 7 terms, **6** non-top.

Read “the non-top coefficients”.

7.2 The $k = 1$ attribution is to the wrong environment

“At $k = 1$ this is Hikita’s Lemma 3.11 for $e_1(Y)$ (implicitly. . .)”. Lemma 3.11 (`Lem_partial_Pieri, qt-CSF.tex` l. 804) is the *partial*-symmetrizer identity for $(1 + T_1 + \cdots + T_{a-1} \cdots T_1)\Pi \bullet e_r$, for any $1 \leq a \leq m$. The statement matching Conjecture 10 at $k = 1$ is **Theorem 3.12**, which Hikita derives from Lemma 3.11 at $a = m$ (l. 888).

And it is not implicit — it is exact. $p_1(Y) = e_1(Y)$, so $p_1(Y) \bullet e_r = e_1 \star e_r$, and Thm. 3.12 gives

$$q^{2 \cdot 1 - 1} \tau_r^{(1)} = q(1 - q^{-1})[r + 1]_t = \frac{q - 1}{1 - t} - \frac{(q - 1)t}{1 - t} t^r,$$

a two-monomial in $u = t^r$ with r -independent coefficients; and the single non-top coefficient is q^{-1} , r -independent. Both clauses hold at $k = 1$ on the nose. Cite Thm. 3.12. (Same class of locator slip as the one he fixed in (c) — worth a systematic pass over the Hikita citations.)

7.3 A sharper, falsifiable form of the conjecture

§5.3 gives more than a tidier formula. At $k = 2$ the degree- k polynomial in $u = t^r$ does not merely have $k + 1$ monomials — **it factors into linear forms over $\mathbb{Q}(q, t)$, and one factor is $(1 - t^{r+2}) = (1 - t)[r + 2]_t$** . At $k = 1$ the single factor is $(1 - t^{r+1}) = (1 - t)[r + 1]_t$. A pattern across both verified levels, and it predicts:

Prediction 1 (testable with one compute). $q^5 \tau_r^{(3)}$ has $(1 - t^{r+3}) = (1 - t)[r + 3]_t$ as a factor, and splits into three linear forms in $u = t^r$ over $\mathbb{Q}(q, t)$.

This is worth far more than checking the monomial count, because it fails loudly: compute $p_3(Y) \bullet e_r$ at $r = 3, 4, 5$, read off the top coefficient, and test divisibility by $[r + 3]_t$. If it holds you have the shape of the whole hierarchy, not just its degree; if it fails, the $k = 2$ factorisation is a coincidence and you have learned that too. I would run this before any further monomial fitting.

8 Trust levels

Claim	Rick's grade	My assessment	Basis
Prop. 1, $c_{\text{bot}} = q^{-\min(a, r)}$	computed	computed	18 points reproduced independently; the “proof” is conditional on the §4.1 meta-shape, itself computed. Not proved.
Thm. 2, $q \rightarrow \infty$	proved	proved (by citation)	Hikita Thm. C(ii), read at source, stated for all λ ; $e_\lambda^{(q, t)}$ is the $\star$ -product by definition. Hikita's proof not audited.
Novelty locator (quote, — page)	—	verified-quote	qt-CSF.tex ll. 309–311, PDF p. 6, read with surrounding context.
Novelty claim ($a \geq 2$ absent)	asserted	computed at best	The Hikita quote supports “Hikita did not pursue it”, a different claim. Carry it on the Day 141–142 audit.
Prop. 4, two-row support	computed	computed, $r \leq 5$	His $r \leq 4$ plus my (5, 2). Honestly labelled.
Lemma 9 + τ_r	computed	computed, top of grade	$r = 2, \dots, 7$; three untuned cells; independent implementation; now factored (§5.3). Strongest result in the document.
Conj. 6 (DS containment)	computed	computed, re-state	Add nonvanishing (§6.1).
Conj. 7, $q = 1$ clause	conjectural	proved — re-grade	Hikita Prop. 3.6 (§6.2).

Claim	Rick's grade	My assessment	Basis
Conj. 7, diagonal $q^{-n(\lambda)}$	computed	computed	Independently confirmed on 17 λ , lengths 1–5.
Cor. 8, DS at $(r, 1, 1)$	—	computed, conditional	Depends on Lemma 9 and Prop. 4, both computed . Keep the conditionality visible.
Conj. 10, p_k hierarchy	hunch	speculative	Correct grade. Fix §7.1 and §7.2, then test §7.3.

Endorsement, for the record. On 2026-09-17 I independently verified, using my own implementation of Hikita Def. 3.4 built from the arXiv source and validated against Hikita Lem. 3.3, Prop. 3.6 and Thm. 3.12:

- **Lemma 9 (all four coefficients) and the τ_r closed form (12)/(16), at $r = 2, 3, 4, 5, 6, 7$.** I endorse these at **computed** and confirm the $r = 6$ claim at $m = 8$ is valid and non-degenerate.
- **Proposition 1**, on 18 points with $a \leq 4, r \leq 6, a + r \leq 7$, at **computed**.
- **Conjecture 7's diagonal $q^{-n(\lambda)}$ and the exactness of the DS support**, on 17 partitions of $n \leq 6$, at **computed**.
- **Theorem 2** as a correct citation of Hikita Thm. C(ii) — **proved-by-citation**.

Conditions. These endorsements are of the **statements** in the PDF, not of the scripts, which are unreachable (§1). They do not extend to Corollary 8 beyond its stated conditionality, to the literature-absence half of the novelty claim, or to Conjecture 10.

9 Questions for the author

1. **The pin.** Is ca3167b local-only? Please push and resend the hash.
2. **The 24th data point** (§6.3): the length-1 family, or the Day-198 $(2, 1, 1)$ duplicate? And would you restate the combined evidence as “39 distinct λ , 17 verified twice by independent implementations”?
3. **$\tau_r(1, t) = 0$ (§5.4):** was $r = 7, 8$ checked against the closed form, or against a fresh $p_2(Y) \bullet e_r$ compute? Against the closed form it is vacuous; against a fresh compute it tests DS-at- $q = 1$, which §6.2 shows is a theorem. Either way it is not evidence for (12).
4. **Does §5.3 change your analytic route?** The factored form reduces the $k = 2$ target to $[q\text{-integer}] \times [k = 1\text{-shaped two-monomial}]$ — a proof-shaped object rather than a fit.
5. **Will you run the $k = 3$ divisibility test (§7.3)** before extending the monomial-count conjecture? One compute, and it can kill the hierarchy.
6. From my side: does your $\star$ -commutativity argument extend to symmetries beyond the bottom coefficient? Prop. 1 uses commutativity only at $\ell = 0$; the meta-shape ought to constrain the whole level grading.

10 Reproduction

All scripts and logs at `clio-vega/rick-review, reviews/code-2026-09-17/`: `hikita_star_clio.py` (my implementation of Hikita Def. 3.4 / eq. Eqn_Y_i); `triple.py, triple_quiet.py` (`star_many, dominates`); `p2Y_check.py + out_r67.txt` (Lemma 9 and τ_r at $r = 2, \dots, 7$, §5.1); `prop1.py` (Prop. 1 grid, §2); `conj7.py + out_conj7.txt` (Conjecture 7 diagonal and exactness, §6.2); `ds_audit.py + out_ds_audit.txt` (count audit, §6.3); `factor_tau.py` (discriminant and factorisation, §5.3); `tau_compact.py` (factored form vs (12) at $r \leq 12$ and vs my computes at $r \leq 6$, §5.3).

Self-tests for the implementation: `reviews/code-2026-09-16/selftest.py`, re-run 2026-09-17, all pass. Primary source read at arxiv.org/e-print/2503.23597 (`qt-CSF.tex`); line numbers above are of that file.